

Khushi Kaushik

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Education

University of California, San Diego

September 2025 - June 2027

Master's in Science, Computer Science

- Natural Language Processing, Advanced Compiler Design, Software Engineering, Advanced Data Bases, Probabilistic AI, New Venture Design

California State University, Fullerton

August 2021 - May 2025

Bachelor of Science, Computer Science (Magna Cum Laude)

Fullerton, CA

- Coursework: C, C++, Python, Operating Systems, Networking, Algorithms, Compilers, Databases, Machine Learning, System Programming

Skills

Systems & Infrastructure: C, Docker, Railway, Multi-threading, Git/GitHub, Linux, Postman

Languages and Backend: C++, Rust, Python, SQL

Frontend & Data: React, Tableau, Plotly, Spark

Applied ML: LangGraph, PyTorch, Scikit-learn, CNNs, Claude

Experience

Thirdsight.ai

June 2025 - Present

SWE Intern La Jolla, CA

- Designed and shipped a validation-focused backend pipeline integrating three LLM providers, building internal tools and automated tests to reproduce edge cases and verify natural-language-to-SQL translation across 40+ scenarios.
- Cut query latency by designing a caching layer with automated result validation and retry logic for ambiguous or failing inputs.
- Built end-to-end pipelines that turn parsed intent into structured queries, surfacing analytics on claim profitability, formulary strategy, and revenue drivers.
- Implemented CSRF and session-based auth against a Railway-hosted REST API for secure production integration.

451 Degrees

February 2026 - Present

AI Engineer La Jolla, CA

- Architected and debugged a horizontally scalable content-detection pipeline (Redis, multi-threaded request handling), reproducing failures under high-traffic conditions and validating fixes through sprint-based regression testing.
- Improved pirated-link detection 200% during live streaming events, using sprint performance analysis and confusion matrices to fix a scoring inversion bug in the detection algorithm.
- Built automated test coverage with Pytest and Jest and added structured logging across the pipeline, improving reliability and reducing debugging time.
- Designed the team's foundational codebase and architecture, and led sprint planning across engineering cycles.

Apple

October 2025 - Present

Part-Time Retail Specialist (Product Zone)

La Jolla, CA

- Diagnosed complex iOS/macOS hardware/software issues for 20+ clients daily under strict time constraints.

Biomechanics Analysis of Human Motion via PA

August 2025 - October 2025

Machine Learning Researcher @ McCulloch Lab UCSD

La Jolla, CA

- Built Python preprocessing pipelines to extract biomechanical markers from multi-angle video, applying PCA to reduce feature dimensionality while preserving key movement signals.
- Benchmarked participant movement metrics against reference models to quantify biomechanical deviations for exercise-form assessment.

SoCal Data Science Research Program

June 2024 - August 2024

Data Science Research Intern

Irvine, CA

- Cut pipeline runtime 40% across 10,000+ simulation frames through targeted Python/Scikit-learn performance tuning and feature selection.
- Fine-tuned regression models to predict structural drift and identify mechanical stress points, reaching $R^2 = 0.947$.
- Architected automated preprocessing pipelines (Python, NumPy) so distributed research teams could reproduce results consistently.

Technical Projects

Inclarity inclarity.onrender.com

Full-stack INCI ingredient normalizer

- Built and deployed Inclarity, a full-stack INCI normalization engine (Python/Flask + vanilla JS) that maps messy cosmetic ingredient strings to canonical names via exact, fuzzy (RapidFuzz), and optional semantic matching
- Designed a 3-tier confidence-routing system (Accept / Review / Unmatched); on a 114-item messy ground-truth set (lexical mode), achieved 99.1% overall accuracy with a 0.0% false-accept rate and ~0.03 ms average lookup latency
- Shipped end-to-end on Render; expanded the in-memory reference from a small sample to 175 INCI entries across 50+ functional categories

HubCaf hubcaf.com

Full-stack cafe discovery platform for remote workers

- Built a full-stack app (TypeScript, pnpm workspaces) with a React 19 frontend and Express 5 API backed by PostgreSQL/Drizzle ORM
- Designed a type-safe API layer using OpenAPI + Orval codegen, auto-generating React Query hooks and Zod schemas shared across frontend/backend
- Implemented authentication with Clerk, interactive maps with Leaflet, and a component system using Radix UI/shadcn/ui + Tailwind CSS v4
- Deployed and shipped to real users, who contribute cafe reviews on wifi, outlets, parking, and noise level

Biomarker Analysis for Neuro-cognitive Decline Post-Cardiac Surgery

Division of Biology and Medicine Brown Center for Biomedical Informatics

- Built an automated feature selection and preprocessing framework for clinical time-series data, cutting pipeline latency 50%.
- Trained and tuned Logistic Regression and XGBoost models to predict neuro-cognitive decline post-cardiac surgery.
- Integrated SHAP to explain feature importance, improving model debuggability and clinician trust.

Compiler Backend: Register Allocation & Garbage Collection

- Built a graph-coloring register allocator in Rust, mapping variables to x86-64 registers/stack slots via conflict-graph analysis, with spilling and save/restore logic to preserve correctness under register pressure
- Implemented a mark-compact garbage collector for heap-allocated vectors — stack-root traversal, forwarding-address compaction, and reference relocation — using a two-word (mark bit + length) object header
- Enforced heap-size limits by tracking live data through GC, correctly triggering OOM errors on over-allocation